

Project Design Phase-II

Data Flow Diagram & User Stories - Auto Ticket Classification System

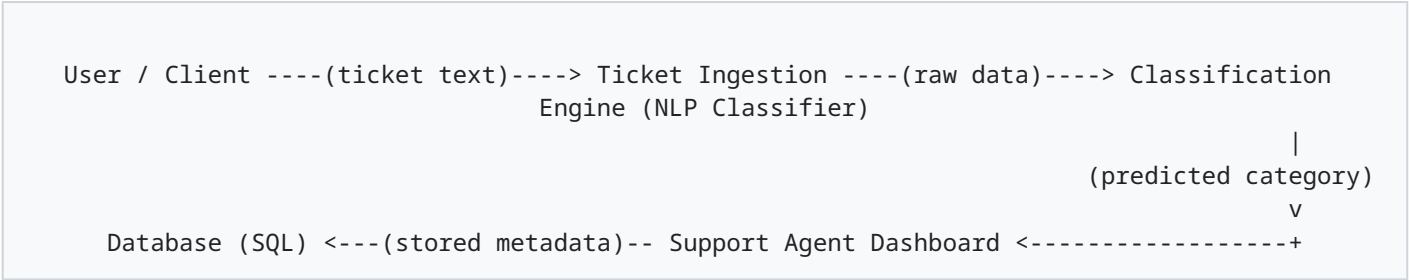
Date	01 October 2026	Maximum Marks	4 Marks
Team ID	SWTID-2026-1536	Project Name	Auto Ticket Classification Using Flow Design

1. Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

System Processing Flow:

1. User / Client submits a support ticket with subject and description text through the Web or Mobile Portal.
2. Ticket Ingestion Module receives incoming ticket requests and validates the payload format.
3. NLP Engine (BERT / TF-IDF Classifier) analyzes the ticket text, performs tokenization, feature extraction, and predicts category, priority, and department.
4. Database / Ticket Repository stores the classified ticket details along with confidence scores and routing history.
5. Dashboard & Notification Service routes the ticket to the respective support agent and visualizes statistics in real-time.



2. User Stories

Below is the detailed list of user stories categorized by user type, functional requirements (epics), priorities, and release sprints for the Auto Ticket Classification System.

User Type	Functional Requirement (Epic)	No.	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web / Mobile user)	Ticket Submission	USN-1	As a user, I can submit a new support ticket by providing a title, category description, and attachment.	Ticket is successfully created with a unique tracking ID and confirmation message.	High	Sprint-1
Customer (Web / Mobile user)	Ticket Status Tracking	USN-2	As a user, I can view the real-time status, predicted category, and assigned agent of my submitted tickets.	Dashboard lists all user tickets with accurate status badges (Open, In Progress, Resolved).	High	Sprint-1
Customer (Web / Mobile user)	Email Notification	USN-3	As a user, I will receive an automated email notification when my ticket is classified and assigned.	Email is received containing ticket summary, predicted category, and tracking link.	Medium	Sprint-2
Support Agent	Queue Management	USN-4	As a support agent, I can view incoming tickets auto-sorted by predicted category and urgency priority.	Agent sees prioritized ticket queue filtered by department and severity.	High	Sprint-1
Support Agent	Classification Feedback	USN-5	As a support agent, I can correct misclassified ticket categories to improve the ML model accuracy over time.	Submitting correction updates database and triggers model retraining pipeline log.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	No.	User Story / Task	Acceptance Criteria	Priority	Release
Administrator	Model Performance Dashboard	USN-6	As an admin, I can monitor classification accuracy metrics, precision, recall, and daily ticket volumes.	Analytics dashboard displays interactive charts showing ML model performance and queue stats.	High	Sprint-2
Administrator	User & Rule Management	USN-7	As an admin, I can configure custom keyword rules and fallback routing policies for edge-case tickets.	Custom rules take immediate effect in routing pipeline without application restart.	Low	Sprint-3